CSE 232
Computer Networks
Homework-2

Group -15

Anushk Kumar	2023115
Ayush Kitnawat	2023160
Abhas Gupta	2023017
Akshat Singh	2023064
Arhan Jain	2023118

Question 1:

Exploring Network Interfaces with ifconfig (5 Marks)

Run the ifconfig command on your machine.

• Identify all available network interfaces (wired, wireless, loopback, etc.).

• For your active interface, report:

– IP address (IPv4 and IPv6 if available)

– MAC address (if available)

– RX/TX packet counts and errors

• Add a screenshot of ifconfig's output.

Solution

Network Interfaces

Wired	en0, en1, en2, en3, bridge0, gif0, stf0
Wireless	awdl0, llw0, anpi0, anpi1, ap1
Loopback	lo0

IP Address (IPv4) (en0) : 192.168.43.165

IP Address (IPv6) (en0) : fe80::894:49ce:df22:31d3%en0

MAC Address (en0) : a4:cf:99:57:15:8d

RX Packet Count : 34401029

TX Packet Count : 8511033

Terminal Shell Edit View Window Help

anushkkumar — anushkkuma

```
Last login: Tue Aug 26 12:44:13 on ttys000
➜  ~ ifconfig
lo0: flags=8049<UP,LOOPBACK,RUNNING,MULTICAST> mtu 16384
        options=1203<RXCSUM,TXCSUM,TXSTATUS,SW_TIMESTAMP>
        inet 127.0.0.1 netmask 0xff000000
        inet6 ::1 prefixlen 128
        inet6 fe80::1%lo0 prefixlen 64 scopeid 0x1
        nd6 options=201<PERFORMNUD,DAD>
gif0: flags=8010<POINTOPOINT,MULTICAST> mtu 1280
stf0: flags=0<> mtu 1280
anpi0: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
        options=400<CHANNEL_IO>
        ether e2:94:2c:62:c5:8f
        media: none
        status: inactive
anpi1: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
        options=400<CHANNEL_IO>
        ether e2:94:2c:62:c5:90
        media: none
        status: inactive
en3: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
        options=400<CHANNEL_IO>
        ether e2:94:2c:62:c5:6f
        nd6 options=201<PERFORMNUD,DAD>
        media: none
        status: inactive
en4: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
        options=400<CHANNEL_IO>
        ether e2:94:2c:62:c5:70
        nd6 options=201<PERFORMNUD,DAD>
        media: none
        status: inactive
en1: flags=8963<UP,BROADCAST,SMART,RUNNING,PROMISC,SIMPLEX,MULTICAST> mtu 1500
        options=460<TSO4,TSO6,CHANNEL_IO>
        ether 36:28:d8:d9:72:00
        media: autoselect <full-duplex>
        status: inactive
en2: flags=8963<UP,BROADCAST,SMART,RUNNING,PROMISC,SIMPLEX,MULTICAST> mtu 1500
        options=460<TSO4,TSO6,CHANNEL_IO>
        ether 36:28:d8:d9:72:04
        media: autoselect <full-duplex>
        status: inactive
bridge0: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
        options=63<RXCSUM,TXCSUM,TSO4,TSO6>
```

Terminal Shell Edit View Window Help

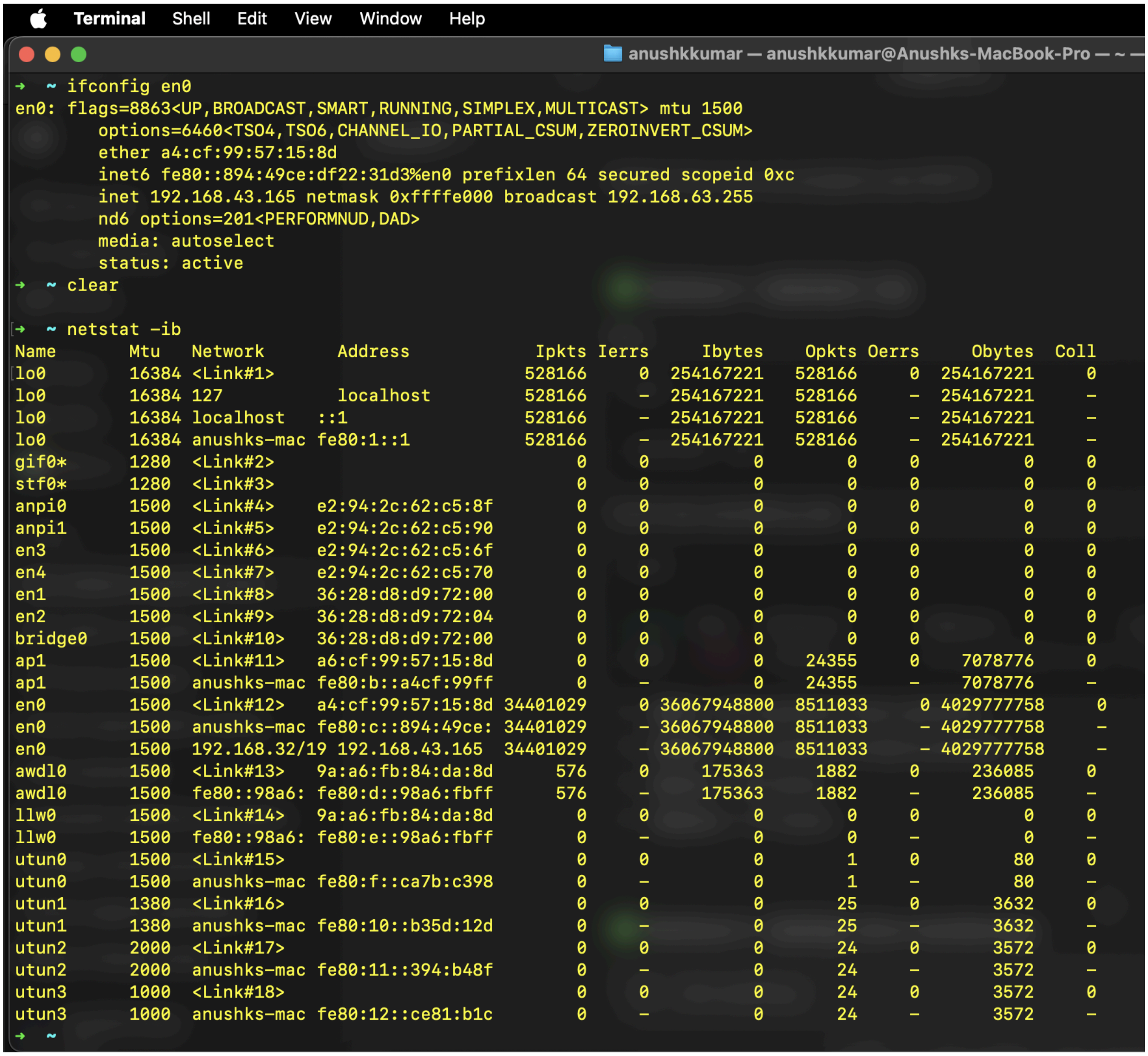
anushkkumar — anush

```
bridge0: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
        options=63<RXCSUM,TXCSUM,TSO4,TSO6>
        ether 36:28:d8:d9:72:00
        Configuration:
                id 0:0:0:0:0:0 priority 0 hellotime 0 fwddelay 0
                maxage 0 holdcnt 0 proto stp maxaddr 100 timeout 1200
                root id 0:0:0:0:0:0 priority 0 ifcost 0 port 0
                ipfilter disabled flags 0x0
        member: en1 flags=3<LEARNING,DISCOVER>
                ifmaxaddr 0 port 8 priority 0 path cost 0
        member: en2 flags=3<LEARNING,DISCOVER>
                ifmaxaddr 0 port 9 priority 0 path cost 0
        nd6 options=201<PERFORMNUD,DAD>
        media: <unknown type>
        status: inactive
ap1: flags=8843<UP,BROADCAST,RUNNING,SIMPLEX,MULTICAST> mtu 1500
        options=6460<TSO4,TSO6,CHANNEL_IO,PARTIAL_CSUM,ZEROINVERT_CSUM>
        ether a6:cf:99:57:15:8d
        inet6 fe80::a4cf:99ff:fe57:158d%ap1 prefixlen 64 scopeid 0xb
        nd6 options=201<PERFORMNUD,DAD>
        media: autoselect (<unknown type>)
        status: inactive
en0: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
        options=6460<TSO4,TSO6,CHANNEL_IO,PARTIAL_CSUM,ZEROINVERT_CSUM>
        ether a4:cf:99:57:15:8d
        inet6 fe80::894:49ce:df22:31d3%en0 prefixlen 64 secured scopeid 0xc
        inet 192.168.43.165 netmask 0xffffe000 broadcast 192.168.63.255
        nd6 options=201<PERFORMNUD,DAD>
        media: autoselect
        status: active
awdl0: flags=8843<UP,BROADCAST,RUNNING,SIMPLEX,MULTICAST> mtu 1500
        options=6460<TSO4,TSO6,CHANNEL_IO,PARTIAL_CSUM,ZEROINVERT_CSUM>
        ether 9a:a6:fb:84:da:8d
        inet6 fe80::98a6:fbff:fe84:da8d%awdl0 prefixlen 64 scopeid 0xd
        nd6 options=201<PERFORMNUD,DAD>
        media: autoselect
        status: active
llw0: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
        options=400<CHANNEL_IO>
        ether 9a:a6:fb:84:da:8d
        inet6 fe80::98a6:fbff:fe84:da8d%llw0 prefixlen 64 scopeid 0xe
        nd6 options=201<PERFORMNUD,DAD>
        media: autoselect
        status: inactive
```

Terminal Shell Edit View Window Help

anushkkumar — anushkk

```
        media: <unknown type>
        status: inactive
ap1: flags=8843<UP,BROADCAST,RUNNING,SIMPLEX,MULTICAST> mtu 1500
        options=6460<TSO4,TSO6,CHANNEL_IO,PARTIAL_CSUM,ZEROINVERT_CSUM>
        ether a6:cf:99:57:15:8d
        inet6 fe80::a4cf:99ff:fe57:158d%ap1 prefixlen 64 scopeid 0xb
        nd6 options=201<PERFORMNUD,DAD>
        media: autoselect (<unknown type>)
        status: inactive
en0: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
        options=6460<TSO4,TSO6,CHANNEL_IO,PARTIAL_CSUM,ZEROINVERT_CSUM>
        ether a4:cf:99:57:15:8d
        inet6 fe80::894:49ce:df22:31d3%en0 prefixlen 64 secured scopeid 0xc
        inet 192.168.43.165 netmask 0xffffe000 broadcast 192.168.63.255
        nd6 options=201<PERFORMNUD,DAD>
        media: autoselect
        status: active
awdl0: flags=8843<UP,BROADCAST,RUNNING,SIMPLEX,MULTICAST> mtu 1500
        options=6460<TSO4,TSO6,CHANNEL_IO,PARTIAL_CSUM,ZEROINVERT_CSUM>
        ether 9a:a6:fb:84:da:8d
        inet6 fe80::98a6:fbff:fe84:da8d%awdl0 prefixlen 64 scopeid 0xd
        nd6 options=201<PERFORMNUD,DAD>
        media: autoselect
        status: active
llw0: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
        options=400<CHANNEL_IO>
        ether 9a:a6:fb:84:da:8d
        inet6 fe80::98a6:fbff:fe84:da8d%llw0 prefixlen 64 scopeid 0xe
        nd6 options=201<PERFORMNUD,DAD>
        media: autoselect
        status: inactive
utun0: flags=8051<UP,POINTOPOINT,RUNNING,MULTICAST> mtu 1500
        inet6 fe80::ca7b:c398:9464:ff75%utun0 prefixlen 64 scopeid 0xf
        nd6 options=201<PERFORMNUD,DAD>
utun1: flags=8051<UP,POINTOPOINT,RUNNING,MULTICAST> mtu 1380
        inet6 fe80::b35d:12d0:198f:1022%utun1 prefixlen 64 scopeid 0x10
        nd6 options=201<PERFORMNUD,DAD>
utun2: flags=8051<UP,POINTOPOINT,RUNNING,MULTICAST> mtu 2000
        inet6 fe80::394:b48f:40d2:9e16%utun2 prefixlen 64 scopeid 0x11
        nd6 options=201<PERFORMNUD,DAD>
utun3: flags=8051<UP,POINTOPOINT,RUNNING,MULTICAST> mtu 1000
        inet6 fe80::ce81:b1c:bd2c:69e%utun3 prefixlen 64 scopeid 0x12
        nd6 options=201<PERFORMNUD,DAD>
➜  ~ _
```



```
 Terminal  Shell  Edit  View  Window  Help
anushkkumar — anushkkumar@Anushks-MacBook-Pro — ~ —
→ ~ ifconfig en0
en0: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
        options=6460<TSO4,TSO6,CHANNEL_IO,PARTIAL_CSUM,ZEROINVERT_CSUM>
        ether a4:cf:99:57:15:8d
        inet6 fe80::894:49ce:df22:31d3%en0 prefixlen 64 secured scopeid 0xc
        inet 192.168.43.165 netmask 0xffffe000 broadcast 192.168.63.255
        nd6 options=201<PERFORMNUD,DAD>
        media: autoselect
        status: active
→ ~ clear

→ ~ netstat -ib
Name       Mtu   Network       Address            Ipkts Ierrs     Ibytes    Opkts Oerrs     Obytes  Coll
lo0        16384 <Link#1>                         528166     0  254167221   528166     0  254167221     0
lo0        16384 127           localhost          528166     -  254167221   528166     -  254167221     -
lo0        16384 localhost   ::1                  528166     -  254167221   528166     -  254167221     -
lo0        16384 anushks-mac fe80:1::1            528166     -  254167221   528166     -  254167221     -
gif0*      1280  <Link#2>                              0     0          0        0     0          0     0
stf0*      1280  <Link#3>                              0     0          0        0     0          0     0
anpi0      1500  <Link#4>    e2:94:2c:62:c5:8f         0     0          0        0     0          0     0
anpi1      1500  <Link#5>    e2:94:2c:62:c5:90         0     0          0        0     0          0     0
en3        1500  <Link#6>    e2:94:2c:62:c5:6f         0     0          0        0     0          0     0
en4        1500  <Link#7>    e2:94:2c:62:c5:70         0     0          0        0     0          0     0
en1        1500  <Link#8>    36:28:d8:d9:72:00         0     0          0        0     0          0     0
en2        1500  <Link#9>    36:28:d8:d9:72:04         0     0          0        0     0          0     0
bridge0    1500  <Link#10>   36:28:d8:d9:72:00         0     0          0        0     0          0     0
ap1        1500  <Link#11>   a6:cf:99:57:15:8d         0     0          0    24355     0    7078776     0
ap1        1500  anushks-mac fe80:b::a4cf:99ff         0     -          0    24355     -    7078776     -
en0        1500  <Link#12>   a4:cf:99:57:15:8d 34401029     0 36067948800  8511033     0 4029777758     0
en0        1500  anushks-mac fe80:c::894:49ce: 34401029     - 36067948800  8511033     - 4029777758     -
en0        1500  192.168.32/19 192.168.43.165  34401029     - 36067948800  8511033     - 4029777758     -
awdl0      1500  <Link#13>   9a:a6:fb:84:da:8d       576     0     175363     1882     0     236085     0
awdl0      1500  fe80::98a6: fe80:d::98a6:fbff       576     -     175363     1882     -     236085     -
llw0       1500  <Link#14>   9a:a6:fb:84:da:8d         0     0          0        0     0          0     0
llw0       1500  fe80::98a6: fe80:e::98a6:fbff         0     -          0        0     -          0     -
utun0      1500  <Link#15>                             0     0          0        1     0         80     0
utun0      1500  anushks-mac fe80:f::ca7b:c398         0     -          0        1     -         80     -
utun1      1380  <Link#16>                             0     0          0       25     0       3632     0
utun1      1380  anushks-mac fe80:10::b35d:12d         0     -          0       25     -       3632     -
utun2      2000  <Link#17>                             0     0          0       24     0       3572     0
utun2      2000  anushks-mac fe80:11::394:b48f         0     -          0       24     -       3572     -
utun3      1000  <Link#18>                             0     0          0       24     0       3572     0
utun3      1000  anushks-mac fe80:12::ce81:b1c         0     -          0       24     -       3572     -
→ ~
```

In Apple Devices, used in this question, the RX, TX packet count is shifted to NetStat Command rather than displaying it on ifconfig.

Sources

https://discussions.apple.com/thread/135456?utm_source=chatgpt.com&sortBy=rank

Question 2:

Tracing Routes with traceroute and mtr (5 Marks)

Use both traceroute and mtr command to trace the route to *google.co.in*.

• Add a screenshot to report for both commands.

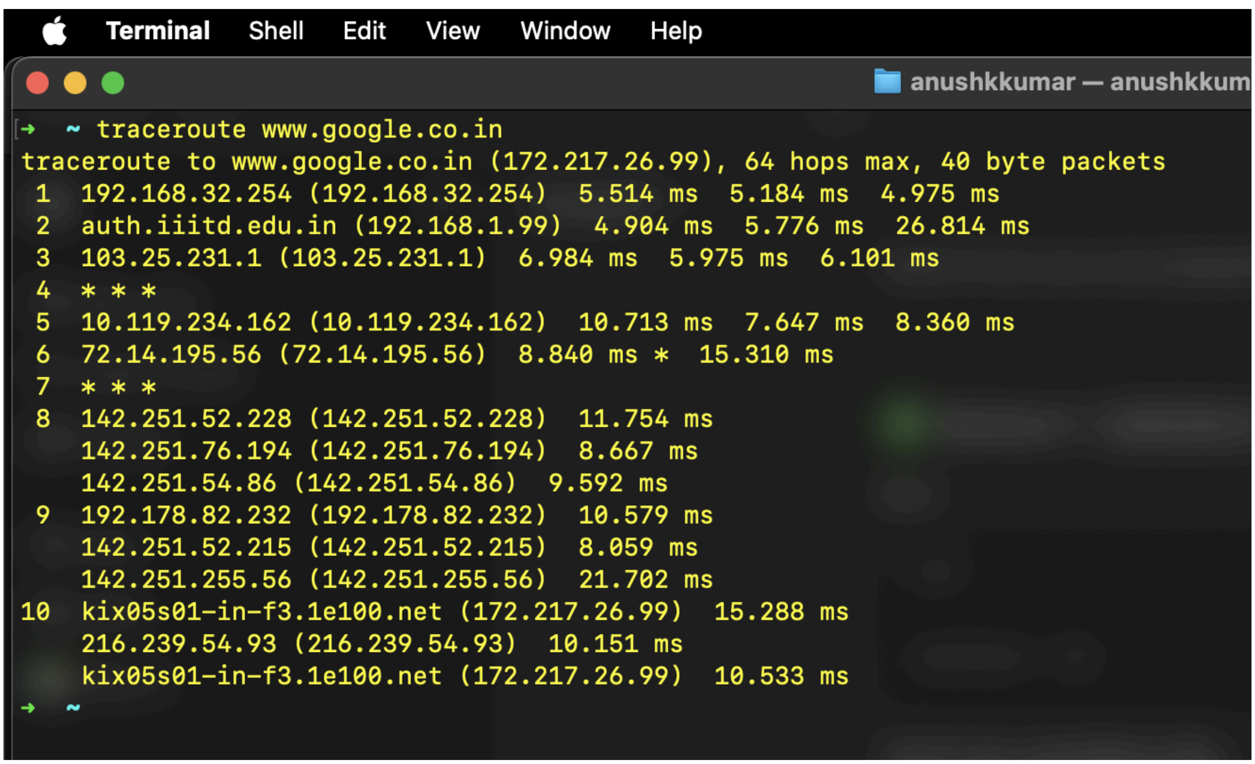
• How many hops or intermediate IP addresses do you observe in each command ?

• If you observe "* * *" in a hop in traceroute's output, then research and report a few possible reasons for it?

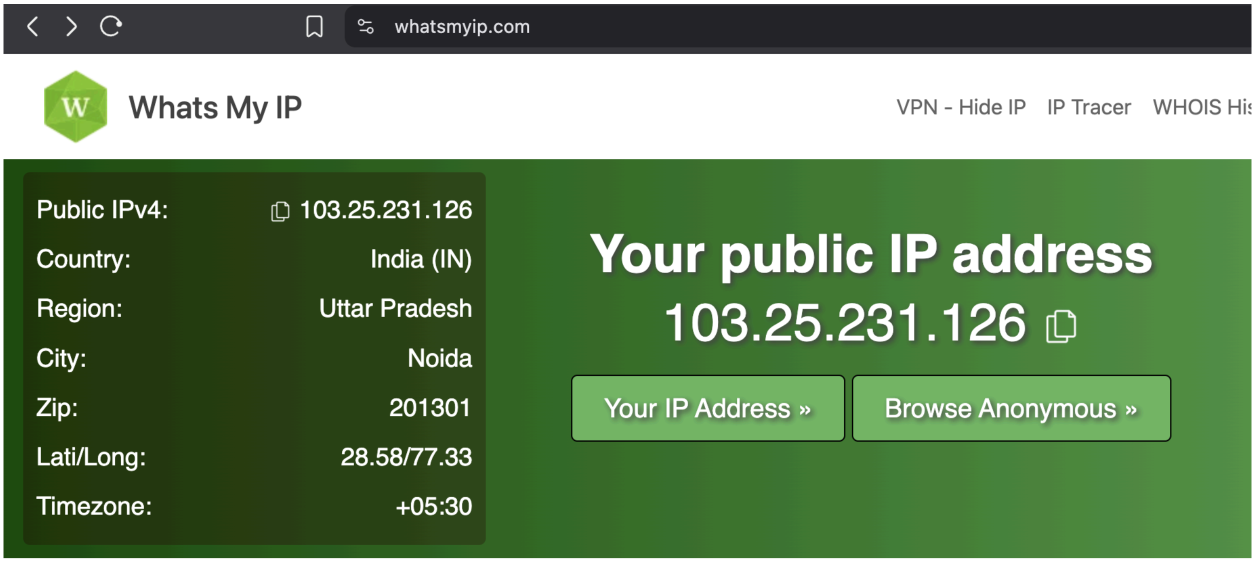
• Lookup your IP address and geolocation using "www.whatismyip.com" website and report it.

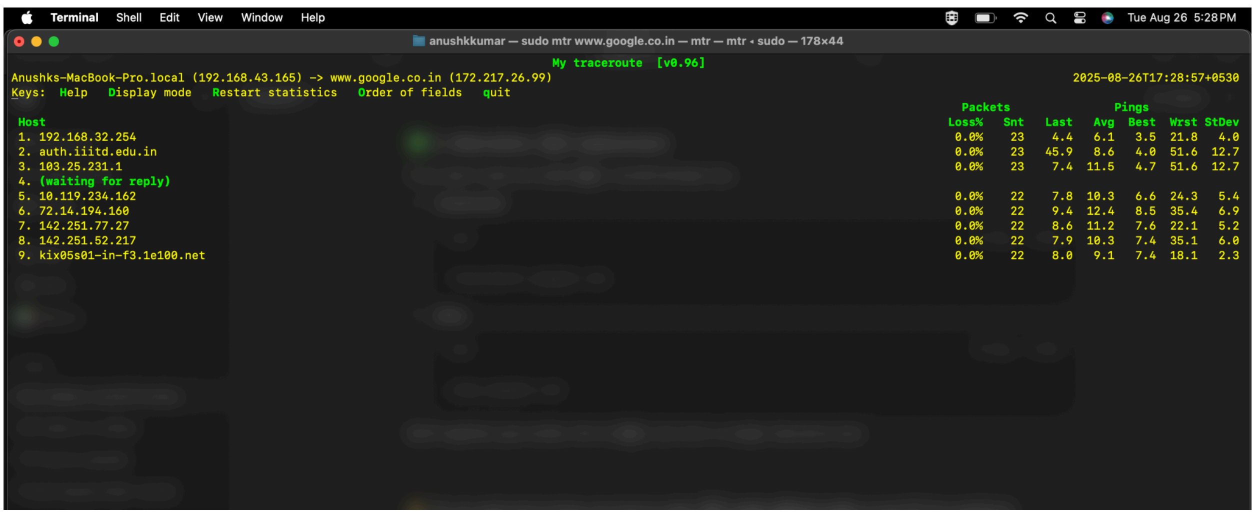
• If there are any non-local IP addresses in the traceroute or mtr, i.e. IP addresses not in ranges 10.0.0.0 to 10.255.255.255, 172.16.0.0 to 172.31.255.255, 192.168.0.0 to 192.168.255.255, then report their geolocation using "check-host.net" website, and the netname, descr attributes using whois shell command. In case traceroute doesn't show any non-local IP address or geolocation isn't available then mention that in the report.

Solution



```
Terminal  Shell  Edit  View  Window  Help
anushkkumar — anushkkum
→ ~ traceroute www.google.co.in
traceroute to www.google.co.in (172.217.26.99), 64 hops max, 40 byte packets
 1  192.168.32.254 (192.168.32.254)  5.514 ms  5.184 ms  4.975 ms
 2  auth.iiitd.edu.in (192.168.1.99)  4.904 ms  5.776 ms  26.814 ms
 3  103.25.231.1 (103.25.231.1)  6.984 ms  5.975 ms  6.101 ms
 4  * * *
 5  10.119.234.162 (10.119.234.162)  10.713 ms  7.647 ms  8.360 ms
 6  72.14.195.56 (72.14.195.56)  8.840 ms *  15.310 ms
 7  * * *
 8  142.251.52.228 (142.251.52.228)  11.754 ms
    142.251.76.194 (142.251.76.194)  8.667 ms
    142.251.54.86 (142.251.54.86)  9.592 ms
 9  192.178.82.232 (192.178.82.232)  10.579 ms
    142.251.52.215 (142.251.52.215)  8.059 ms
    142.251.255.56 (142.251.255.56)  21.702 ms
10  kix05s01-in-f3.1e100.net (172.217.26.99)  15.288 ms
    216.239.54.93 (216.239.54.93)  10.151 ms
    kix05s01-in-f3.1e100.net (172.217.26.99)  10.533 ms
→ ~
```





- In **traceroute** command **10** hops, and in **mtr** command **9** hops

- * * * represents no ICMP signal received on that particular hop/RTT cycle. Reasons for the same could be :
  - Network Congestion
  - Packet Loss at Gateway
  - Security Policies & Firewall restricting the network to reveal the identity of the host server.

- The Global IP Address through whatsmyip.com is 103.25.231.126, GeoLoc : Noida, Uttar Pradesh, IN

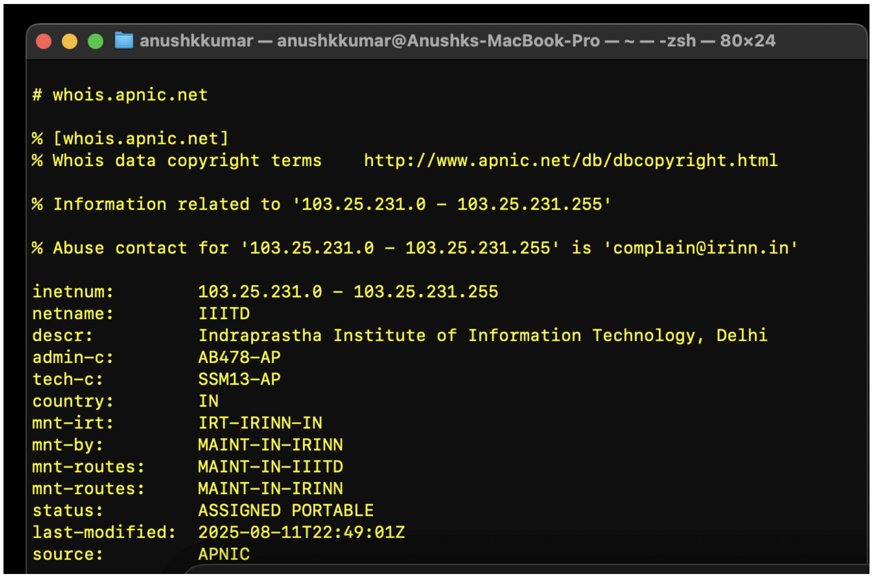
- Non-Local IP Address

Traceroute hops

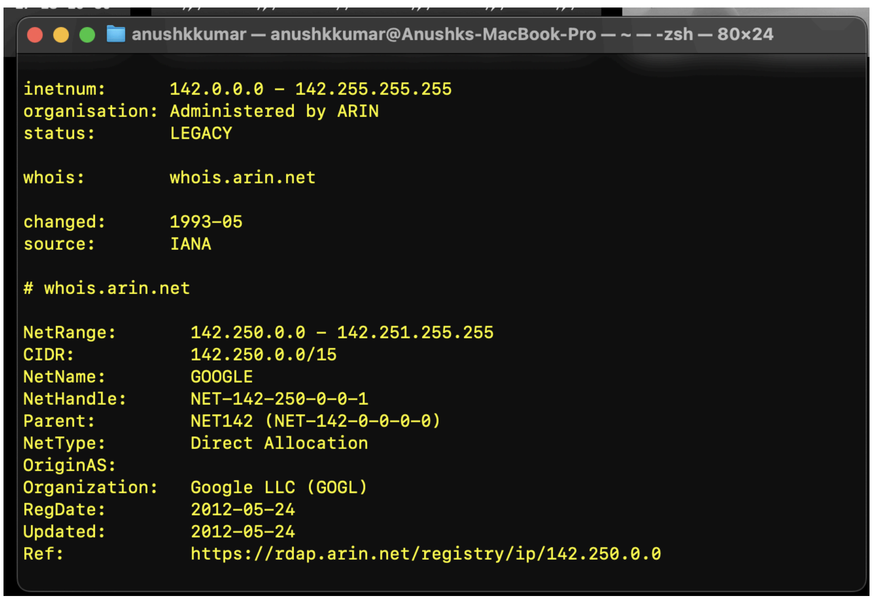
IP Addresses	*Owner*	*Type*	*GeoLoc*
103.25.231.1	IIIT Delhi		New Delhi,IN
72.14.195.56	Google LLC	BackBone	California,US
142.251.52.228 / 142.251.76.194 / 142.251.54.86	Google LLC	BackBone	Chennai,IN
192.178.82.232	Google LLC	G-CDN	Chennai,IN
142.251.52.215 / 142.251.255.56	Google LLC	BackBone	Chennai,IN
216.239.54.93	Google LLC	GlobalCache	California,US
172.217.26.99	Google LLC	India Server	Chennai,IN

Mtr Hops

IP Addresses	*Owner*	*Type*	*GeoLoc*
103.25.231.1	IIIT Delhi		Delhi,IN
10.119.234.162	Data not found	Private Network	
72.14.194.160	Google LLC	BackBone	Chennai,IN
142.251.77.27	Google LLC	G-CDN	Chennai,IN
142.251.52.217	Google LLC	BackBone	Chennai,IN
172.217.26.99	Google LLC	India Server	Chennai,IN




```
# whois.apnic.net

% [whois.apnic.net]
% Whois data copyright terms    http://www.apnic.net/db/dbcopyright.html

% Information related to '103.25.231.0 - 103.25.231.255'

% Abuse contact for '103.25.231.0 - 103.25.231.255' is 'complain@irinn.in'

inetnum:        103.25.231.0 - 103.25.231.255
netname:        IIITD
descr:          Indraprastha Institute of Information Technology, Delhi
admin-c:        AB478-AP
tech-c:         SSM13-AP
country:        IN
mnt-irt:        IRT-IRINN-IN
mnt-by:         MAINT-IN-IRINN
mnt-routes:     MAINT-IN-IIITD
mnt-routes:     MAINT-IN-IRINN
status:         ASSIGNED PORTABLE
last-modified:  2025-08-11T22:49:01Z
source:         APNIC
```




```
inetnum:      142.0.0.0 - 142.255.255.255
organisation: Administered by ARIN
status:       LEGACY

whois:        whois.arin.net

changed:      1993-05
source:       IANA

# whois.arin.net

NetRange:       142.250.0.0 - 142.251.255.255
CIDR:           142.250.0.0/15
NetName:        GOOGLE
NetHandle:      NET-142-250-0-0-1
Parent:         NET142 (NET-142-0-0-0-0)
NetType:        Direct Allocation
OriginAS:
Organization:   Google LLC (GOGL)
RegDate:        2012-05-24
Updated:        2012-05-24
Ref:            https://rdap.arin.net/registry/ip/142.250.0.0
```

Question 3:

DNS Analysis with nslookup and dig (5 Marks)

Use nslookup and dig to query the IP addresses of google.com.

• Compare the outputs of both tools. Do they report the same IP address?
  When you use this IP address as URL in browser, does it open Google's homepage?

• Find IP address of an Authoritative Nameserver of "google.com" using command "nslookup -type=NS google.com". Then query the authority name server using command "nslookup google.com – ⟨IP address of authoritative name server ⟩". Attach screenshot of this step in the report.

Solution:

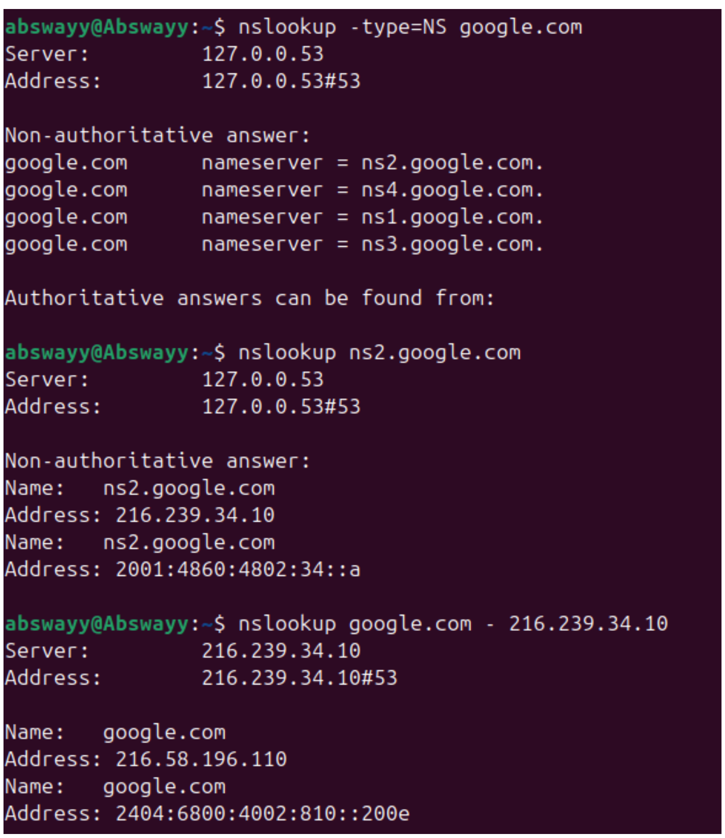
```
abswayy@Abswayy:~$ nslookup google.com
Server:         127.0.0.53
Address:        127.0.0.53#53

Non-authoritative answer:
Name:   google.com
Address: 142.250.194.14
Name:   google.com
Address: 142.250.193.110
Name:   google.com
Address: 142.250.182.174
Name:   google.com
Address: 172.217.26.110
Name:   google.com
Address: 172.217.26.46
Name:   google.com
Address: 2404:6800:4002:81d::200e
```

```
abswayy@Abswayy:~$ dig google.com

; <<>> DiG 9.18.30-0ubuntu0.24.04.2-Ubuntu <<>> google.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 58390
;; flags: qr rd ra; QUERY: 1, ANSWER: 5, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;google.com.                    IN      A

;; ANSWER SECTION:
google.com.             250     IN      A       142.250.182.174
google.com.             250     IN      A       142.250.194.14
google.com.             250     IN      A       142.250.193.110
google.com.             250     IN      A       172.217.26.46
google.com.             250     IN      A       172.217.26.110

;; Query time: 1 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Wed Aug 27 18:51:35 IST 2025
;; MSG SIZE  rcvd: 119
```

- **Do they report the same IP address?**
  Yes, both the commands (nslookup and dig) returns the same IP addresses just in a different order. All output addresses are the same, but nslookup returns an additional 2404:6800:4002:81d::200e address, which is an IPv6 address (source:https://tools.tracemyip.org/lookup/2404:6800:4002:81d::200e), that the dig command did not return.

- **When you use this IP address as URL in browser, does it open Google's homepage?**
  Yes, upon typing all the IP addresses into Firefox's search bar, it opened google.com, except the extra IPv6 address 2404:6800:4002:81d::200e returned by nslookup (typing it in the search bar does not load into google.com homepage).

- **Find IP address of an Authoritative Nameserver of "google.com" using command "nslookup -type=NS google.com". Then query the authority name server using command "nslookup google.com – 〈IP address of authoritative name server 〉". Attach screenshot of this step in the report**
  The authoritative nameserver used for this question: ns2.google.com
  First we returned the names of authoritative nameservers of google.com using nslookup -type=NS google.com, then chose one of the authoritative nameservers and grabbed it's IP address using nslookup again, and then used the command in question to query the authority name server.



```
absswayy@Absswayy:~$ nslookup -type=NS google.com
Server:         127.0.0.53
Address:        127.0.0.53#53

Non-authoritative answer:
google.com      nameserver = ns2.google.com.
google.com      nameserver = ns4.google.com.
google.com      nameserver = ns1.google.com.
google.com      nameserver = ns3.google.com.

Authoritative answers can be found from:

absswayy@Absswayy:~$ nslookup ns2.google.com
Server:         127.0.0.53
Address:        127.0.0.53#53

Non-authoritative answer:
Name:   ns2.google.com
Address: 216.239.34.10
Name:   ns2.google.com
Address: 2001:4860:4802:34::a

absswayy@Absswayy:~$ nslookup google.com - 216.239.34.10
Server:         216.239.34.10
Address:        216.239.34.10#53

Name:   google.com
Address: 216.58.196.110
Name:   google.com
Address: 2404:6800:4002:810::200e
```

Question 4:

Measuring Traffic with iftop (5 Marks)

Run iftop while browsing or streaming content on your browser.

• Add a screenshot of the top 3 connections consuming bandwidth to the report.

• Report the source/destination IPs and bandwidth usage.

• What type of applications might be generating this traffic?

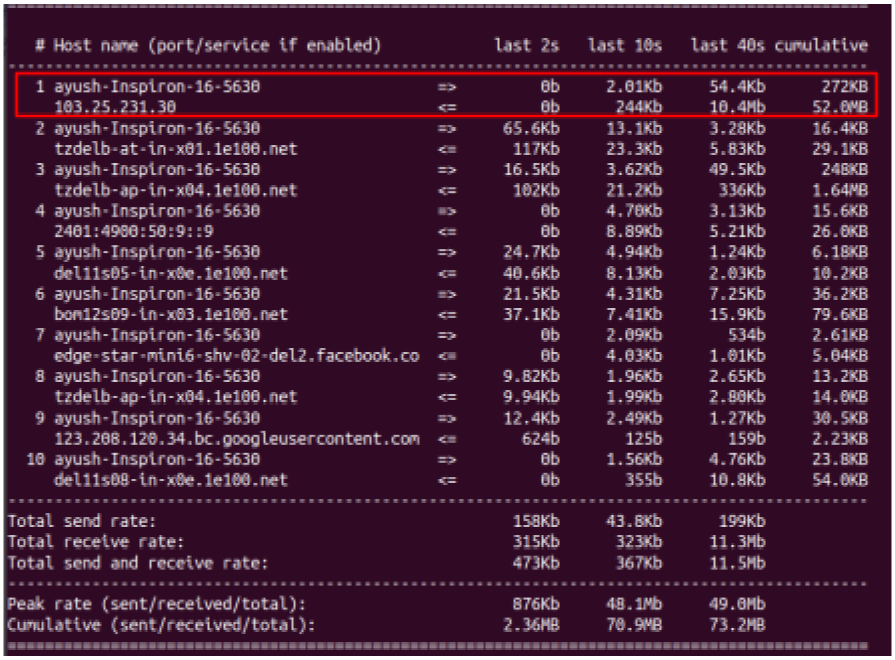
<u>Solution</u>

“=>”: This arrow indicates that the data is sent from x host to y host like.(x⇒y)(uploading data)

“<=”: This arrow means the data is received from y host to x host like.(x⇐y)(downloading data)

In the below screenshots, we are monitoring my network interface over Wifi where the host is **ayush-inspiron-16-5630**. The red highlighted ones are the top connections consuming bandwidth.

```
=========================================================================================

  # Host name (port/service if enabled)            last 2s   last 10s   last 40s cumulative
-----------------------------------------------------------------------------------------
  1 ayush-Inspiron-16-5630                    =>     159Kb     31.8Kb     7.96Kb     39.8KB
    cdn-185-199-111-154.github.com            <=    7.20Mb     1.44Mb      369Kb     1.80MB
  2 ayush-Inspiron-16-5630                    =>    24.4Kb     58.7Kb     44.7Kb      320KB
    tzdelb-ap-in-x04.1e100.net                <=    4.05Kb      298Kb      253Kb     2.00MB
  3 ayush-Inspiron-16-5630                    =>        0b     10.7Kb     2.68Kb     13.4KB
    lcmaaa-al-in-x0e.1e100.net                <=        0b      155Kb     38.8Kb      194KB
  4 ayush-Inspiron-16-5630                    =>        0b     3.74Kb     1.13Kb     21.7KB
    bom12s19-in-x03.1e100.net                 <=        0b      145Kb     36.7Kb      295KB
  5 ayush-Inspiron-16-5630                    =>    63.2Kb     12.6Kb     3.16Kb     15.8KB
    20.207.73.82                              <=     340Kb     67.9Kb     17.0Kb     84.9KB
  6 ayush-Inspiron-16-5630                    =>        0b     29.0Kb     7.26Kb     36.3KB
    pnbomb-ba-in-x0e.1e100.net                <=        0b     50.5Kb     12.6Kb     63.1KB
  7 ayush-Inspiron-16-5630                    =>        0b     10.1Kb     2.52Kb     12.6KB
    del11s18-in-x0a.1e100.net                 <=        0b     51.1Kb     12.8Kb     63.9KB
  8 ayush-Inspiron-16-5630                    =>        0b     8.63Kb     2.16Kb     10.8KB
    lcmaaa-ax-in-x0e.1e100.net                <=        0b     45.5Kb     11.4Kb     56.9KB
  9 ayush-Inspiron-16-5630                    =>        0b     20.3Kb     5.06Kb     25.3KB
    tzdelb-ap-in-x04.1e100.net                <=        0b     21.2Kb     5.30Kb     26.5KB
 10 ayush-Inspiron-16-5630                    =>        0b     10.1Kb     2.53Kb     12.6KB
    tzdelb-bj-in-x0e.1e100.net                <=        0b     23.2Kb     5.80Kb     29.0KB
-----------------------------------------------------------------------------------------
Total send rate:                                     277Kb      242Kb      215Kb
Total receive rate:                                 7.60Mb     2.35Mb     11.7Mb
Total send and receive rate:                        7.87Mb     2.58Mb     11.9Mb
-----------------------------------------------------------------------------------------
Peak rate (sent/received/total):                     876Kb     48.1Mb     49.0Mb
Cumulative (sent/received/total):                   2.66MB     73.8MB     76.4MB
=========================================================================================
```



#	Host name (port/service if enabled)		last 2s	last 10s	last 40s	cumulative
1	ayush-Inspiron-16-5630	=>	0b	2.01Kb	54.4Kb	272KB
	103.25.231.30	<=	0b	244Kb	10.4Mb	52.0MB
2	ayush-Inspiron-16-5630	=>	65.6Kb	13.1Kb	3.28Kb	16.4KB
	tzdelb-at-in-x01.1e100.net	<=	117Kb	23.3Kb	5.83Kb	29.1KB
3	ayush-Inspiron-16-5630	=>	16.5Kb	3.62Kb	49.5Kb	248KB
	tzdelb-ap-in-x04.1e100.net	<=	102Kb	21.2Kb	336Kb	1.64MB
4	ayush-Inspiron-16-5630	=>	0b	4.70Kb	3.13Kb	15.6KB
	2401:4900:50:9::9	<=	0b	8.89Kb	5.21Kb	26.0KB
5	ayush-Inspiron-16-5630	=>	24.7Kb	4.94Kb	1.24Kb	6.18KB
	del11s05-in-x0e.1e100.net	<=	40.6Kb	8.13Kb	2.03Kb	10.2KB
6	ayush-Inspiron-16-5630	=>	21.5Kb	4.31Kb	7.25Kb	36.2KB
	bom12s09-in-x03.1e100.net	<=	37.1Kb	7.41Kb	15.9Kb	79.6KB
7	ayush-Inspiron-16-5630	=>	0b	2.09Kb	534b	2.61KB
	edge-star-mini6-shv-02-del2.facebook.co	<=	0b	4.03Kb	1.01Kb	5.04KB
8	ayush-Inspiron-16-5630	=>	9.82Kb	1.96Kb	2.65Kb	13.2KB
	tzdelb-ap-in-x04.1e100.net	<=	9.94Kb	1.99Kb	2.80Kb	14.0KB
9	ayush-Inspiron-16-5630	=>	12.4Kb	2.49Kb	1.27Kb	30.5KB
	123.208.120.34.bc.googleusercontent.com	<=	624b	125b	159b	2.23KB
10	ayush-Inspiron-16-5630	=>	0b	1.56Kb	4.76Kb	23.8KB
	del11s08-in-x0e.1e100.net	<=	0b	355b	10.8Kb	54.0KB
	Total send rate:		158Kb	43.8Kb	199Kb	
	Total receive rate:		315Kb	323Kb	11.3Mb	
	Total send and receive rate:		473Kb	367Kb	11.5Mb	
	Peak rate (sent/received/total):		876Kb	48.1Mb	49.0Mb	
	Cumulative (sent/received/total):		2.36MB	70.9MB	73.2MB	

Network 1

Source IP	:	ayush-Inspiron-16-5630
Destination IP	:	tzdelb-ap-in-x04.1e100.net
Sent cumulative	:	320KB
Received cumulative	:	2.0MB
Application	:	**Running Youtube**

Bandwidth Calculation:

- Sent: 320 KB = 320 x 1024 bytes × 8 = **2621440** bits.

  Bandwidth =2621440 / 60 = 43690.67 bits/s = **43.7** Kb/s.

- Received: 2.00 MB = 2 x 1024 x 1024 bytes × 8 = **16777216** bits.

  Bandwidth = 16777216 / 60 = 279620.27 bits/s = **279.6** Kb/s.

- Total: 43.7 Kb/s + 279.6 Kb/s = **323.3** Kb/s.

Network 2

Source IP	:	ayush-Inspiron-16-5630
Destination IP	:	cdn-185-199-111-154.github.com
Sent cumulative	:	39.8KB
Received cumulative	:	1.80MB
Application	:	**Browsing github repository**

Bandwidth Calculation:

- Sent: 39.8 KB = 39.8 x 1024 bytes × 8 = **326041.6** bits.

  Bandwidth = 326041.6/ 60 = 5434.027 bits/s = **5.43** Kb/s.

- Received: 1.80 MB = 1.8 x 1024 x 1024 bytes × 8 = **15099494.4** bits.

  Bandwidth = 15099494.4/ 60 = 251658.24 bits/s = **251.67** Kb/s.

- Total: 5.43 Kb/s + 251.67 Kb/s = **257.1 Kb**/s.

Network - 3

Source IP	:	ayush-Inspiron-16-5630
Destination IP	:	103.25.231.30(IIITD server)
Sent cumulative	:	272KB
Received cumulative	:	52.0MB
Application	:	**Accessed IIITD website for TT**

Bandwidth Calculation:

- Sent: 272 KB = 272 x 1024 bytes × 8 = **2228224** bits.

  Bandwidth = 2228224/60 = 37137.066 bits/s = **37.1** Kb/s.

- Received: 52.0 MB = 52 x 1024 x1024 bytes × 8 = **436207616** bits.

  Bandwidth = 436207616 / 60 =7270126.93 bits/s = **7.27** Mb/s.

- Total: 37.1 Kb/s + 7.27 Mb/s = **7.3** Mb/s.